

CALM VERITAS

The Folate (Vitamin B9) Guide

Deficiency, MTHFR, and Dosing —
What the Research Actually Supports

Why Folate Matters

Folate (vitamin B9) is required for DNA synthesis, red blood cell formation, and the methylation cycle that regulates gene expression and homocysteine levels. It works as a coenzyme partner to vitamin B12 — deficiency in either can produce the same megaloblastic anemia, which is why isolated folate testing without B12 testing can be misleading.

Who Is Most at Risk

— Women of childbearing age

The CDC has recommended 400mcg of folic acid daily since 1992 because neural tube defects form in the first 3-4 weeks of pregnancy, often before a person knows they are pregnant. Mandatory folic acid fortification of U.S. grain products began in 1998 and was followed by a documented 19-46% drop in neural tube defect rates across affected regions, per CDC surveillance data.

— People with an MTHFR gene variant

The C677T variant of the MTHFR gene, carried by an estimated 30-40% of some populations in one or two copies, reduces the enzyme's ability to convert folate into its active form, L-methylfolate, by up to 70% in homozygous carriers. This does not mean folic acid is unsafe for these individuals, but it is the basis for interest in pre-methylated folate supplements.

— Heavy alcohol users

Alcohol interferes with folate absorption in the small intestine and increases urinary folate excretion, compounding risk in populations that also tend to have lower dietary intake.

Folate vs. Folic Acid vs. Methylfolate

These three terms are often used interchangeably in marketing but are not identical. Folate is the umbrella term for the vitamin as it occurs naturally in leafy greens, legumes, and liver. Folic acid is the fully oxidized synthetic form used in fortified foods and most standard supplements because it is stable and cheap to produce — but it requires multiple enzymatic conversion steps, including the one affected by MTHFR variants, before the body can use it. L-methylfolate (5-MTHF) is the pre-converted, bioactive form that bypasses the MTHFR-dependent step entirely.

A 2014 review in the British Journal of Pharmacology found L-methylfolate produced measurably higher blood folate levels than equivalent doses of folic acid in individuals carrying the C677T variant, though for the general population without the variant, standard folic acid remains well absorbed and effective.

The Masking Risk

High-dose folic acid supplementation can correct the anemia caused by B12 deficiency while doing nothing to stop the neurological damage B12 deficiency also causes — the anemia symptom disappears while nerve damage continues silently. This is the primary reason the U.S. Institute of Medicine set a tolerable upper intake level of 1,000mcg/day of synthetic folic acid for adults, a ceiling that does not apply to folate from food or L-methylfolate supplements.

Dosing

The RDA for adults is 400 micrograms of dietary folate equivalents (DFE) daily. Pregnant individuals are advised to take 600mcg DFE, and the CDC recommendation of 400mcg folic acid specifically for women capable of pregnancy is meant to be layered on top of normal dietary intake, not as a replacement for it. Supplemental doses above 1,000mcg of folic acid should generally be avoided outside of clinical supervision, given the B12-masking concern above.

Food Sources Worth Knowing

— Leafy greens

Spinach, romaine, and turnip greens are among the densest natural folate sources — a cup of cooked spinach provides roughly 15% of the DFE RDA.

— Legumes

Lentils are unusually concentrated: one cooked cup provides close to 90% of the adult RDA.

— Liver

Beef liver is one of the single richest sources of natural folate per serving, alongside its high B12 and vitamin A content.

Quick Reference

- Adult RDA

400mcg DFE/day from food and supplements combined.

- Pregnancy RDA

600mcg DFE/day; supplementation should begin before conception when possible.

- Upper limit (synthetic folic acid)

1,000mcg/day for adults, per the Institute of Medicine — does not apply to food folate or methylfolate.

- Consider methylfolate if

You carry a known MTHFR C677T variant, or have not responded to standard folic acid supplementation despite low folate status.

- Always pair with B12 testing

Folic acid alone can mask a B12 deficiency, so isolated folate supplementation without B12 status is not advised for anyone with anemia symptoms.